Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – VI (Batch 2024-2028)

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 6	OE	24BT321T	Molecular Diagnostics	3	0	0	3	3
	PE	24BT334T	Biomaterial and implants	3	0	0	3	3
		24BT335T	Novel Separation Processes – Dyes and Pigments					
		24BT336T	Cheminformatics and Medicinal chemistry					
	PE	24BT337T	Pharmacokinetics and Pharmaco Dynamics	3	0	0	3	3
		24BT338T	Food Biotechnology					
		24BT339T	Next Generation Sequence Analysis					
	PC	24BT304T	Genetic Engineering techniques	3	0	0	3	3
	PC	24BT304P	Genetic Engineering techniques Laboratory	0	0	2	2	1
	PC	24BT305T	Tissue Engineering and Regenerative Medicine	3	0	0	3	3
	PC	24BT205P	Tissue Engineering and Regenerative Medicine Laboratory	0	0	2	2	1
	PC	24BT306T	Pharmacology of Drug Action	3	0	0	3	3
	PC	24BT306P	Pharmacology of Drug Action Laboratory	0	0	2	2	1

24BT321T					Molecular Diagnostics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept on transmission of infection and ethical and legal aspects of diagnostics.
2. To introduce pathogenicity and diagnosis.
3. To study different techniques related to diagnosis.
4. To understand the concepts and types of molecular diagnosis.

UNIT I: INTRODUCTION 10 Hrs.

Infection –mode of transmission in infections, factors predisposing to microbial pathogenicity, types of infectious diseases. Philosophy and general approach to clinical specimens, Sample collection- method of collection, transport and processing of samples. Interpretation of results. Host-Parasite relationships. Quality, Ethical, and legal implication of diagnostics, international standards, NABH standards, FDA regulation of molecular diagnostics

UNIT II: MICROBIAL PATHOGENICITY 10 Hrs.

Pathogenicity and diagnosis of infection caused by Streptococcus, Coliforms, Salmonella, Shigella, Vibrio and Mycobacterium. Diagnosis of fungal infections. Major fungal diseases: Dermatophytosis, Candidiosis and Aspergillosis.~~Pathogen~~

UNIT III: DIAGNOSTIC TECHNIQUES 10 Hrs.

Diagnosis of DNA and RNA viruses. Pox viruses, Adenoviruses, Rhabdo Viruses, Hepatitis Viruses and Retroviruses. Diagnosis of Protozoan diseases: Amoebiosis, Malaria, trypanosomiasis, Leishmaniasis. Diagnosis of helminthic diseases Fasciola hepatica and Ascaris lumbricoides. Filariasis and Schistosomiasis

UNIT IV: MOLECULAR DIAGNOSIS 10 Hrs.

Human disorders: Biochemical disorders, Immune disorders, chromosomal disorders and single cell disorders. Chromosomal disorder diagnosis -autosomal; sex chromosomal; karyotype analysis. G-banding, in situ hybridization (FISH and on-FISH), and comparative genomic hybridization (CGH). Cancer cytogenetics Spectral karyotyping Genes in pedigree. Genetic Counselling.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify different types of infectious diseases and the ethics in diagnosis
CO2 : Describe methods of sample collection, transport, and processing, including interpretation of result
CO3 : Interpret diagnostic test results to identify specific microbial pathogens.
CO4 : Compare and contrast the clinical features of different fungal diseases to guide differential diagnosis.
CO5 : Discuss ethical considerations in the collection, handling, and analysis of clinical specimens.
CO6 : Critically appraise the regulatory framework for diagnostic testing and propose enhancements to ensure patient safety and quality of care.

TEXT/REFERENCE BOOKS

1. Nader Rifai, A. Rita Horvath, Carl T. Wittwer, Jason Park, “Principles and Applications of Molecular Diagnostics - A Derivative of the Tietz Textbook of Clinical Chemistry and Molecular Diagnostics” 6th ed.
2. Betty A. Forbes , Daniel F. Sahm, Alice S. Weissfeld , Ernest A. Trevino, “ Bailey & Scott’s Diagnostic Microbiology” (2002), Published by C.V. Mosby.
3. Geo F. Brooks, Stephen A. Morse, Janet S. Butel, “ Jawetz, Melnick, & Adelberg’s Medical Microbiology” (2004)
4. David E. Bruns, Edward R. Ashwood, Carl A. Burtis, “Fundamentals of Molecular Diagnostics” (2007).. Saunders Group

24BT334T					Biomaterial & Implants					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals of Biomaterial & Implants
2. To understand the basic concept of bio materials that are used in medical devices.
3. To acquire the knowledge on Bio-materials
4. To compare the testing protocols of wide range of Bio implants and devices

UNIT I: INTRODUCTION TO BIOMATERIALS 10 Hrs.

Biomaterial, Types of Biomaterials, Biocompatibility, Biological material, Biodegradable material, Bioresorbable material, Bio-inert material, Bio-active material, Minimum Requirements of Biomaterials, Surface Properties of Biomaterials, Desirable Properties of Biomaterial

UNIT II: DIFFERENT TYPES OF BIOMATERIALS 8 Hrs.

Different types of metallic and polymeric biomaterials. Applications and Performance of Biomaterials. Impact of material corrosion and its prevention strategies.

UNIT III: BIOCOMPATIBILITY TESTING & RESPONSE OF BIOMATERIAL 12 Hrs.

Biocompatibility Testing: Introduction, *In-Vitro* Testing, *In-Vivo* Testing, Hypersensitivity, Haemocompatibility, Odontocompatibility, Osteocompatibility, Cytotoxicity, Genotoxicity, Carcinogenicity. Response of Biomaterial to Human Body: Blood-Biomaterial Interactions, Biomaterials-Tissue Interactions, Tissue response to Implants, Inflammation, Wound Healing, and Foreign Body Response, Impact of Biofilm formation on biomaterials.

UNIT IV: BIO-IMPLANTS 10 Hrs.

Stent, Vascular grafts, Artificial Heart valves, Contact lenses, Intra-ocular lenses, Artificial Silicon Retina, Temporary fixation Devices, Total Hip Replacement, Total Knee Replacement, Dental filling & Restoration material, Dental implants, Suture materials, Wound dressings, Tissue Adhesives.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Summarise the classifications and properties of different biomaterials, their properties, performance specification andbiological applications

CO2 : Demonstrate the applications of Metallic & Ceramic based biomaterials

CO3 : Illustrate the types and applications of Bio-implants

CO4 : Analyze the essential qualities and requirement of biomaterials for implant fabrication.

CO5 : Compare different testing protocols for specific biomaterial/implant and evaluate the response of biomaterial/Implant on biological environment.

CO6 : Apply the knowledge of biomaterials to judge which material in order to Design and develop suitable Bio-implants.

TEXT/REFERENCE BOOKS

1. Sujata V. Bhatt," Biomaterials", Narosa Publishing House, New Delhi.
2. Buddy D. Ratner, Allan Hoffman, Frederick J. Schoen, Jack E. Lemons," Biomaterials Science: An introduction to materials in medicine", Elsevier.
3. R.Narayan," Biomedical Materials", Springer Science.
4. Joon B. Park, Joseph D. Bronzino," Biomaterials: Principles and Applications", CRC Press.
5. Joon B. Park, Roderic S. Lakes," Biomaterials: An introduction,", Springer.
6. Fredrick H," Biomaterials Medial Devices and Tissue Engineering", Silver Chapman and Hall.
7. J. B. Park "Biomaterials science and engineering", Plenum press, New York.

24BT335T					Novel Separation Processes – Dyes and Pigments					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the pigmentary property, chemistry behind the colorants.
2. To explain its applications in various field according to the chemistry involved.
3. To create awareness on testing of pigments synthesis processing and applications of dyes pigments lakes
4. To narrate basic aspects of separation processes

UNIT I: NOVEL SEPARATION PROCESS 12 Hrs.

Introduction – Need for separation process – Overview of separation processes – types and classification. Membrane processes – choice of membranes- Membrane filtration – micro, nano and ultrafiltration. Dialysis and reverse osmosis. Chromatographic separations –Ion-exchange separation. Liquid and gaseous separations.

UNIT II: POLYMERS 10 Hrs.

Introduction, Classification of Polymers - origin, structure, intermolecular interactions, Types of polymerization – addition, condensation, radical, ionic and copolymerization. Inorganic polymers, organic polymers, biodegradable polymers, Photonic polymers, fire retarding polymers, membranes and smart materials – Introduction, types and applications

UNIT III: DYES AND PIGMENTS 10 Hrs.

Introduction, Nomenclature and Classification of Natural and Synthetic dyes, Color and constitution - chromospheres and auxochromes with suitable examples, Azo dyes - types of azo dyes, synthesis of acidic and basic azo dyes. Natural dyes – structure determination and synthesis of alizarine, Quinazarin and Indigo. Food colours. Introduction to Fluorescence dyes and Introduction to laser dyes. Pigments: Introduction, Structures of Porphyrins, plant pigments and classification, Bile pigments. Synthesis of Haemin and Chlorophyll. Synthetic pigments – preparation of Phthalocyanines..

UNIT IV: REMEDIATION PROCESSES 10 Hrs.

Impact of the textile and leather dye Industry on the environment, Pollution of water with dyes and pigments, Toxicity of food colors, Need for remediation, Health and Environmental Hazards of Synthetic Dyes and their Remediation Processes, Effluent Treatment Strategies for industrial dye removal

42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Able to narrate the separation process
CO2 : Understand the different kinds of polymers
CO3 : Able to understand fundamental knowledge on basics of chemistry involved in the colorants
CO4 : Able to explain the types of dyes on the basis of application, properties.
CO5 : Able to explain the synthetic methods used for azo dyes and their properties.
CO6 : Health care problems and environmental issues related to industrial dyes

TEXT/REFERENCE BOOKS

1. Jens Nielsen, Gregory Stephanopoulos, Sang Yup Lee, “Metabolic Engineering: Concepts and Applications”, Wiley , 2021
2. P. Wankat “Separation Process Engineering”. Fifth Edition. 2022
3. AM Lane. “Separation Process Essentials”. CRC Press 2020.
4. F. W. Billmeyer Jr. “Textbook of Polymer Science” John Wiley & sons , 2017
5. P. S. Vankar “Handbook on Natural Dyes for Industrial Applications”, National Institute of Industrial Research, 2016

24BT336T					Cheminformatics & Medicinal Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To introduce the small molecule-ligand-oriented *in silico* Physico-chemical aspects of rational drug design
2. To Gain insight on chemoinformatics
3. To Introduce Computational Chemistry Techniques in Drug Design
4. To Explore Applications of Chemoinformatics in Drug Discovery

UNIT I: CHEMISTRY & INFORMATION TECHNOLOGY 10 Hrs.

Overview of Rational Drug Design, Ligands and Targets, *in-silico* representation of chemical information. Chemical Databases: Data Mining, Chemical/biochemical data collation, retrieval, analysis & interpretation.

UNIT II: MOLECULAR DRAWING AND INTERACTIVE VISUALIZATION 10 Hrs.

Building molecules on a computer, Molecular Modeling. Computer-Aided Drug Design: Overview, Structural Homology Modeling Tools, Docking Tools, and Screening Tools. Chemical data science: Artificial intelligence in chemistry, simulation methods for molecules and materials

UNIT III: MECHANISM 10 Hrs.

Stereochemistry and mechanism, coordination chemistry for drug design, *in silico* tools for medicinal chemistry (docking, MD, *de novo* drug design), Organic reaction mechanism, Logic in organic synthesis, QSAR, pharmacological screening, chemistry of drug action, Pharmaceutical Preformulation, Solid State Pharmaceutics, Drug metabolism, pharmacokinetics, pharmacodynamics.

UNIT IV: STRUCTURE ACTIVITY RELATIONSHIPS 10 Hrs.

Antihistaminic agents, anti-anginal, vasodilator, calcium channel blocker, Diuretics, Anti-hypertensive agents, Anti-arrhythmic drugs, anti-hyperlipidemic agents, coagulants & anticoagulants, drug used in congestive Heart failure, Drugs acting on Endocrine system, Antidiabetic agents.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Investigate chemicals and materials that are not practical for laboratory analysis

CO2 : Develop individual model molecules or the behaviors of chemical compounds within the natural world; create and/or work with databases to catalog, categorize, organize, and search the structures of chemicals

CO3 : Employ computational chemistry to simplify problems and make calculations that are used in laboratory experimentation

CO4 : Apply computational chemistry techniques and chemoinformatics tools to investigate chemical compounds

CO5 : Synthesize and analyze chemical data using computational methods

CO6 : Evaluate the effectiveness and limitations of computational chemistry methods and chemoinformatics tools in rational drug design

TEXT/REFERENCE BOOKS

1. Muthukumarasamy Karthikeyan and Renu Vyas. “Practical chemoinformatics”. Springer,ISBN 9788132234913, 2014.
2. Silverman, Richard B., and Mark W. Holladay. “The organic chemistry of drug design and drug action”. Academic Press, 2014
3. Bajorath, Jurgen. “Chemoinformatics for Drug Discovery”. John Wiley & Sons, 2013.
4. Cramer, C.J.,” Essentials of Computational Chemistry”, 2nd Ed., John Wiley & Sons Ltd., 2004.
5. Foye. “Essentials of Foye's Principles of Medicinal Chemistry” – 2016.

24BT337T					Pharmacokinetics & Pharmacodynamics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the kinetics of drug absorption, distribution, metabolism, excretion, elimination.
2. To impart the Pharmacodynamics of drug action and drug interactions
3. To analyse various pharmacokinetic & pharmacodynamics parameters, their significance & applications.
4. To compare the Bioavailability and bioequivalence of drug products and their significance.

UNIT I: ABSORPTION AND DISTRIBUTION 10 Hrs.

Absorption - Introduction, Mechanisms of Drug Absorption, Factors Influencing Drug Absorption, Distribution – Factors influencing Distribution of Drug, Volume of Distribution and its significance.

UNIT II: METABOLISM AND EXCRETION 10 Hrs.

Drug Metabolism - Introduction, Drug Metabolism, Metabolic Pathways, Factors Affecting drug metabolism. Excretion - Introduction, Renal Excretion of Drugs, Factors Affecting the Renal Excretion of Drugs, Clearance, Renal Clearance, Non-renal Routes of the Drug Excretion.

UNIT III: PHARMACODYNAMICS & DRUG INTERACTIONS 10 Hrs.

Principles and Mechanism of Drug action, Drug interactions: Effect of protein- binding interactions, the effect of tissue-binding interactions, cytochrome p450-based drug interactions, drug interactions linked to drug-drug, drug-food, drug-alcohol interactions.

UNIT IV: APPLICATION OF PHARMACOKINETICS & DYNAMICS 10 Hrs.

Dose response relationship, Therapeutic index, factors modifying drug action. Modified-Release Drug formulations, Targeted Drug Delivery Systems. Bioavailability and Bioequivalence. Clinical significance of bioavailability and bioequivalence studies.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain fundamental knowledge on Absorption, Distribution, Metabolism and Elimination
CO2 : Understand the key steps involved in pharmacokinetics and Pharmacodynamics
CO3 : Analyze and compare different types of drug interaction and its clinical significance
CO4 : Get acquainted with metabolic pathway and excretion mechanism
CO5 : Focus on Application of Pharmacokinetics in Bioavailability and Bioequivalence
CO6 : Design and develop pharmaceutical formulations and validate its bioavailability and bioequivalence.

TEXT/REFERENCE BOOKS

1. D. M. Brahmankar and Sunil B.Jaiswal, “Bio pharmaceutics and Pharmacokinetics-A Treatise”, Vallabh Prakashan Pitampura, Delhi .
2. Leon Shargel and Andrew B,” Applied biopharmaceutics and pharmacokinetics,” Mc Graw Hill.
3. Remington,”Remington’s Pharmaceutical Sciences”, Mack Publishing Company,.
4. Abdou H.M, “Dissolution, Bioavailability and Bioequivalence” Mack Publishing Company, Pennsylvania.
5. Malcolm Rowland and Thomas, N. Tozen,”Clinical Pharmacokinetics, Concepts and Applications”, Lippincott Williams and Wilkins.
6. Milo Gibaldi, “Biopharmaceutics and Clinical Pharmacokinetics” Pharma Book Syndicate.

24BT338T					Food Biotechnology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the fundamentals of food biotechnology
2. To learn the concepts of food processing and preservation
3. To emphasize the concept of functional foods and different fermented foods
4. To comprehend the principles behind important analytical techniques employed in biotechnology as well as in genetic modification of foods.

UNIT I: FOOD PROCESSING AND PRESERVATION 10 Hrs.

Introduction to food processing of various foods including dairy, bakery, brewing, fruit and vegetable products, plantation products, oilseeds, meat, fish, poultry; pro and prebiotics and nutraceuticals. Principles of food preservation by dehydration, thermal treatments like pasteurization, sterilization, canning, retorting etc., low temperature i.e., chilling and freezing, chemical preservation/ bio-preservation, traditional methods like salting/ syruping, pickling, fermentation etc., non- thermal processes like MAP, irradiation, high pressure processing etc., and hurdle technology

UNIT II: FOOD BIOTECHNOLOGY 10 Hrs.

Fermentative production of enzymes used in food industry; solid state fermentation; recovery of enzymes from natural sources; cheese making and whey processing, impact of enzyme technology (bioethanol, protein hydrolysates, bioactive peptides); enzymatic processing of fruit juices. Role of enzymes in baking, meat and meat processing; comparative methods of toxicity test in (novel) foods; biosensors; enzymatic approach to tailor made fats; catabolic processes and oxygen-dependent reactions in food; use of lipases and reactions in organic solvents and two phases.

UNIT III: FERMENTATION FOOD BIOTECHNOLOGY 10 Hrs.

Fermented foods: Dairy products, oriental fermentations, alcoholic beverages, and food ingredients; the role of biotechnology in fermented food products (dairy, meat, vegetable); Starter culture development, process development; Enzymes in the dairy industry: cheese making and whey processing, impact of enzyme technology; Functional foods. Nutraceuticals, Use of Antibiotics & Hormones in Food Processing & Agricultural Practices etc.

UNIT IV: FOOD SAFETY AND REGULATION 10 Hrs.

Role of WHO to improve evaluation of GM food, Benefits & Controversies, Irradiated Food, Labelling of Irradiated Food. Freeze dried food, Functional Foods & Nutraceuticals, Functional foods from plant sources, animal sources, dietary supplements, Regulation. FAO in India, Technical Cooperation programmes, Biosecurity in Food and Agriculture, World Health Organization (WHO), World Animal Health Organization (OIE), International Plant Protection Convention (IPPC); Codex Alimentarius Commission - Codex India – Role of Codex Contact point, National Codex contact point (NCCP), National Codex Committee of India – ToR, Functions, Shadow Committees etc.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Demonstrate the propagation of plant cultures
CO2 : Perform the maintenance and culturing of animal cell
CO3 : Understand the principles of gene construct
CO4 : Get acquainted with the ethical issues involved in transgenic organisms
CO5 : Focus professionally on latest technologies like genome editing
CO6 : Apply their knowledge on transgenic plants in day to day life

TEXT/REFERENCE BOOKS

1. B. Lee, “Fundamentals of Food Biotechnology”, 2nd Edition, Wiley-Blackwell Publishers, 2020

2. SC Bhatia, “Food Biotechnology (Woodhead Publishing India in Food Science, Technology and Nutrition)”, CRC Press, 2017
3. HD Belitz, W. Grosch, P Schieberle. “Food Chemistry”, Springer, 2009

24BT339T					Next Generation Sequence Analysis					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Learn the specific information of of Next Generation Sequencing
2. Gain the fundamental on the various platforms used in NGS.
3. Create interest in learning the tools and techniques used in NGS data analysis.
4. Learn the applications and scopes of genomics research using the latest genome-wide data centric approaches.

UNIT I: INTRODUCTION 10 Hrs.

Historical perspective of sequencing technologies. Overview of modern sequencing technologies. Next Generation sequencing platforms. Data acquisition and base calling. Quality of sequencing data. Read mapping and genome assembly. Structural and functional annotation of genomes. Statistics and algorithms used in different steps in data analysis.

UNIT II: TECHNOLOGIES FOR TRANSCRIPTOMICS AND REGULATORY GENOMICS 10 Hrs.

Chip- seq Analysis. Peak finding. Motif Discovery. RNA-seq Analysis. Differential gene expression analysis. Exome sequencing and analysis. Exome sequencing.

UNIT III: SOFTWARE AND PIPELINES FOR NGS DATA ANALYSIS 10 Hrs.

Chip-seq analysis Piplines. RNA-seq analysis pipelines. Software used for assembly and differential gene analysis. Basics of Genome Browsers. Annotation pipelines.

UNIT IV: GENOME SEQUENCING AND APPLICATIONS IN GENETICS STUDIES 10 Hrs.

Algorithms and application in studying regulation of gene expression. Emerging technologies of single-cell gene expression analysis. Metagenomics. Variant detection. Time series analysis. Pathway Analysis.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify the different sequencing technologies.
CO2 : Gain the fundamental practice of transcriptomics and genomics.
CO3 : Understand about the pipelines for RNA Seq analysis.
CO4 : List different mapping and genome assembly methods.
CO5 : Focus professionally on software's and tools for NGS data analysis
CO6 : Develop solution to enhance understanding for genome sequencing and genetic studies.

TEXT/REFERENCE BOOKS

1. JM Bower and H Bolouri, eds, "Computational modeling of genetic and biochemical networks", MIT Press 2019
2. Noam Shomrom, "Deep Sequencing Analysis", Springer, 2018
3. Fall, C.P., Marland, E.S., Wagner, J.M., Tyson, J.J, "Computational Cell Biology", Springer, 2018.
4. Megahed Mohammad, "Genomic Data Analysis", LAP Lambert Academic Publishing, 2017
5. Choi Sangdun, "Introduction to System Biology", Humana Press/Trtowa/New Jersey, 2013.

24BT304T					Genetic Engineering Techniques					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/VIVA	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. The objective of the course is to teach basics of theory and practical aspects of recombinant DNA technology
2. Provide theoretical bases for the application of versatile DNA modifying enzymes, cloning strategies, vector types
3. The course will also outline the applications of this knowledge for the development of diagnostics, therapeutics and vaccines.
4. Students will be introduced to prominent nucleic acid labelling techniques

UNIT I: INTRODUCTION 10 Hrs.

Overview of gene cloning. Restriction, modification systems and DNA modifying enzymes, DNA analysis by electrophoresis. Plasmids and bacteriophages as vectors for gene cloning. Cloning vectors for E. coli like pBR322, pUC8, pGEM3Z. Cloning vectors based on M13 and λ bacteriophage. Ti plasmid, BAC and YAC.

UNIT II: INTRODUCTION OF DNA INTO CELLS AND SELECTION OF RECOMBINANTS 10 Hrs.

Ligation of DNA molecules. Introduction of DNA into cells, Transformation, selection for transformed cells. Identification of recombinants, blue-white selection. Identification of recombinant phages. cDNA and Genomic libraries.

UNIT III: RECOMBINANT DNA METHODS 10 Hrs.

Fundamentals of polymerase chain reaction, designing primers for PCR. DNA sequencing by Sanger's method and automated DNA sequencing, Vectors for expression of foreign genes in E. coli, cassettes and gene fusions. Production of recombinant protein by eukaryotic cells. Fusion tags and their role in purification of recombinant proteins.

UNIT IV: TECHNIQUES IN BIOTECHNOLOGY 10 Hrs.

Blotting technique, RT-PCR, In-situ hybridization, RAPD, RFLP, Antisense RNA and Ribozyme technology, CRISPR/Cas9 gene editing, Site-directed mutagenesis and protein engineering; Applications of recombinant DNA technology; Safety regulations related to genetic engineering

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Memorize the Principles and importance of gene cloning
CO2 : Explain various methods for screening of recombinants and identification of cloned gene
CO3 : Apply the methods of Polymerase chain reaction and DNA sequencing.
CO4 : Analyze the application of recombinant technology
CO5 : Evaluate structure and organization of different vectors used in gene transfer
CO6 : Design primers for amplification of genes

TEXT/REFERENCE BOOKS

1. Brown, T. A. (2020), "Gene Cloning and DNA Analysis: An Introduction", (8th ed.)Wiley-Blackwell Publishing (Oxford, UK); ISBN: 978-1-119-07256-0
2. Glick, B.R., Pasternak, J.J., Patten, C. L. "Molecular Biotechnology: Principles and Applications of Recombinant DNA" (4th ed.). ASM Press (Washington DC); ISBN: 978-1-55581- 498-4. (2010)
3. S. B. Primrose, R. Twyman, R.W. Old, "Principles of Gene Manipulation", Wiley-Blackwell; 7th Edition. 2013
4. Bruce Alberts, "Molecular Biology of the cell", 6th edition. New York: Garland Science; 2014
5. Jane K. Setlow, "Genetic Engineering: Principles and Methods", Alexander Hollaender. Volume 6. 2006

24BT304P					Genetic Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. To demonstrate the components required for gene manipulation
2. To apply the knowledge of genetic material and their transformation at molecular and cellular levels
3. To Enhance Critical Thinking and Problem-Solving Skills in Gene Manipulation
4. To Promote Ethical Awareness and Responsibility in Genetic Engineering

LIST OF EXPERIMENTS

1 Making competent *E. coli* cells using chemical methods.
2 Transformation of *E. coli* and calculation of competence.
3 Cloning and expression of GFP sequence in the expression vector
4 Transformation and antibiotic selection or Blue white selection method
5 Restriction digestion with plasmid and lambda Phage DNA
6 Methods of Restriction mapping, using online NEB cutter tools
7 Ligation process (Joining of DNA)
8 In vitro amplification of DNA by PCR or gene pulling
9 DNA elution in agarose gel
10 Agarose gel electrophoresis and visualization using gel doc

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Memorize the Principles and importance of gene cloning
CO2 : Paraphrase various methods to transfer foreign genes
CO3 : Apply concepts of gene cloning principles
CO4 : Analyze experimental data obtained from molecular biology experiments
CO5 : Assess innovative approaches for gene manipulation, recombinant DNA technology
CO6 : Design construct the recombinant vector and develop genetically modified organisms

TEXT/REFERENCE BOOKS

1. Primrose, S.B. and Twyman, R.M," Principles of Gene Manipulation and Genomics", 8th Edition, Blackwell Publishing Co. UK. 2012
2. Brown, T. A. (2020)," Gene Cloning and DNA Analysis: An Introduction", (8th ed.). Wiley-Blackwell Publishing (Oxford, UK); ISBN: 978-1-119-07256-0
3. Christopher Howe," Gene Cloning and Manipulation", 2nd Edition, Cambridge University Press, UK. 2015
4. Ausubel, F.M. et al. (2012),"Current protocols in molecular biology", New York: John Wiley & Sons.
5. Green, M. R., & Sambrook, J. "Molecular cloning: A laboratory manual" (4th ed., Vol. 1-3). Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press. (2012)

<24BT305T					Tissue Engineering & Regenerative Medicine					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. Learn the Basics of Tissue Engineering: Understand tissue structure, repair mechanisms, and cell-matrix interactions essential for tissue engineering.
2. Understand Biomaterials and Their Use: Explore biomaterials' properties, types, and applications in scaffold design and nanotechnology.
3. Gain Knowledge of Stem Cells: Learn about stem cell biology, differentiation, and their therapeutic potential in regenerative medicine.
4. Explore Advances in Tissue Engineering: Study innovative techniques like 3D bioprinting and address safety, ethical, and regulatory aspects.

UNIT I: TISSUE ARCHITECTURE 10 Hrs.

Tissue types and Tissue components, Tissue repair, Engineering wound healing and sequence of events. Basic wound healing. Applications of growth factors: VEGF/angiogenesis, Basic properties, Cell-Matrix & Cell-Cell Interactions, telomeres and Self-renewal, Control of cell migration in tissue engineering.

UNIT II: BIOMATERIALS 10 Hrs.

Biomaterials: Properties of biomaterials, Surface, bulk, mechanical and biological properties. Scaffolds & tissue engineering, Types of biomaterials, biological and synthetic materials, Biopolymers, Applications of biomaterials, Modifications of Biomaterials, Role of Nanotechnology.

UNIT III: BIOLOGY OF STEM CELL 10 Hrs.

Stem Cells: Introduction, hematopoietic differentiation pathway Potency and plasticity of stem cells, sources, embryonic stem cells, hematopoietic and mesenchymal stem cells, Stem Cell markers, FACS analysis, Differentiation, Stem cell systems- Liver, neuronal stem cells, Types & sources of stem cell with characteristics: embryonic, adult, haematopoietic, fetal, cord blood, placenta, bone marrow, primordial germ cells, cancer stem cells induced pleuri potent stem cells.

UNIT IV: CLINICAL APPLICATIONS 10 Hrs.

Stem cell therapy, Molecular therapy, *In vitro* organogenesis, Neurodegenerative diseases, spinal cord injury, heart disease, diabetes, burns and skin ulcers, muscular dystrophy, orthopaedic applications, Stem cells and Gene therapy. 3D bioprinting of tissues and organs. Tissue engineered product characterization, components, safety, efficacy. Preservation –freezing and drying. Patent protection and regulation of tissue-engineered products, ethical issues.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain the fundamental knowledge of tissue architecture.
CO2 : Understand the mechanism of tissue repair.
CO3 : Classify different categories of Biomaterials used in tissue engineering.
CO4 : Get acquainted with stem cell characteristics and their relevance in medicine.
CO5 : Focus on properties and broad applications of biomaterials.
CO6 : Design and develop 3D scaffolds and engineered substitutes for clinical conditions

TEXT/REFERENCE BOOKS

1. Bernhard O.Palsson, Sangeeta N.Bhatia, "Tissue Engineering" Pearson Publishers 2009.
2. Meyer, U.; Meyer, Th.; Handschel, J.; Wiesmann, H.P.Fundamentals of Tissue Engineering and Regenerative Medicine.2009.
3. Bernard N. Kennedy (editor). Stem cell transplantation, tissue engineering, and cancer applications,Nova Science Publishers, 2008.
4. Raphael Gorodetsky, Richard Schäfer. Stem cell-based tissue repair. RSC Publishing, 2011.

24BT305P					Tissue Engineering and Regenerative Medicine Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand techniques involved polymerization
2. To Observe quality control tests of common biomaterials
3. To design appropriate parameters to evaluate the properties of biomaterials
4. To introduce basic concepts of validation of biomaterials

LIST OF EXPERIMENTS

1 To perform polymerization reaction using Self cure resin
2 To fabricate polymer blends
3 To perform physical properties of hardening of metals
4 To perform absorption spectra of ocular biomaterial
5 Fabrication of hydroxyapatite and PEG complexes
6 To analyze the setting temperature of bone cement
7 To analyze the surface texture of the polymers
8 To analyze the tensile strength of the polymers
9 To perform cellulose biomaterial preparation
10 To perform the elctrospun matrix of the blended polymers

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Study the fundamentals of biomaterials and their properties
CO2 : Understand the nature and characteristic features of biomaterials
CO3 : Apply the technology in biomaterial fabrication
CO4 : Analyze the applications of biomaterials
CO5 : Evaluate quality test for biomaterials
CO6 : Design and fabricate bio implants

TEXT/REFERENCE BOOKS

1. Sujata V. Bhatt," Biomaterials", Narosa Publishing House, New Delhi.
2. Buddy D. Ratner, Allan Hoffman, Frederick J. Schoen, Jack E. Lemons," Biomaterials Science: An introduction to materials in medicine", Elsevier.
3. R.Narayan," Biomedical Materials", Springer Science.
4. Joon B. Park, Joseph D. Bronzino," Biomaterials: Principles and Applications", CRC Press.
5. Joon B. Park, Roderic S. Lakes," Biomaterials: An introduction,", Springer.
6. Fredrick H," Biomaterials Medial Devices and Tissue Engineering", Silver Chapman and Hall.
7. J. B. Park "Biomaterials science and engineering", Plenum press, New York.

24BT306T					Pharmacology of Drug Action					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To get acquainted with different categories of drugs and fundamentals of drug action
2. To familiarize the mechanism of drug action of different categories drugs in disease condition
3. To understand the therapeutic uses and adverse effects of drugs across various disease condition
4. To develop the ability to critically evaluate the drug information and to make informed decisions about the therapy.

UNIT I: GENERAL PHARMACOLOGY AND DRUGS ACTING ON CARDIO VASCULAR SYSTEM 10 Hrs.

Receptor theories and classification of receptors, regulation of receptors. Drugs used in congestive heart failure, Anti-anginal drugs, Fibrinolytics and anti-platelet drugs, Anti-hypertensive drugs, Anti-hyperlipidemic drugs.

UNIT II: DRUGS ACTING ON PERIPHERAL NERVOUS SYSTEM AND AUTOCOIDS 10 Hrs.

Skeletal muscle relaxants (peripheral), Local anaesthetic agents and Non-steroidal anti-inflammatory agents, Anti-rheumatic drugs, Anti -asthmatic drugs.

UNIT III: DRUGS ACTING ON CENTRAL NERVOUS SYSTEM 10 Hrs.

Opioid analgesics, centrally acting muscle relaxants, Anti-epileptics, Antidepressants, Anti-anxiety agents, Drugs used in Parkinson's disease and Alzheimer's disease, Alcohols and disulfiram.

UNIT IV: ANTIMICROBIAL AGENTS 12 Hrs.

General principles of chemotherapy, Mechanism of action of Sulfonamides and cotrimoxazole. Penicillins, cephalosporins, chloramphenicol, macrolides, quinolones and fluoroquinolins, tetracycline and aminoglycosides, Antifungal agents, Antiviral drugs.

42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the fundamental principles of drug action and its application in clinical practice.
CO2 : Demonstrate and gain comprehensive knowledge of different categories of drugs acting on the cardiovascular system, peripheral nervous system and autocoids, central nervous system, and chemotherapy.
CO3 : Analyze and compare the mechanism of action, therapeutic uses, and adverse effects of various pharmacological agents.
CO4 : Apply pharmacological principles in the rational selection of drugs for the treatment of various diseases.
CO5 : Evaluate and interpret the pathological processes in disease condition and the role of the pharmacological agents in modulating the process.
CO6 : Investigate and develop safe pharmacological interventions through critical analysis.

TEXT/REFERENCE BOOKS

1. Mycek M.J, Gelnet S.B and Perper M.M. "Lippincott's Illustrated Reviews- Pharmacology", Lippincott Williams and Wilkins.
2. Rang H. P., Dale M. M., Ritter J. M., Flower R. J."Rang and Dale's Pharmacology" ,.Churchil Livingstone Elsevier.
3. Kulkarni SK. "Handbook of experimental pharmacology", Vallabh Prakashan Publisher.
4. Katzung B. G., Masters S. B., Trevor A. J." Basic and clinical pharmacology", Tata Mc Graw-Hill.
5. Bjorn Knollmann, "Goodman and Gilman's The Pharmacological Basis of Therapeutics", McGraw-Hill Education.
6. Sharma H. L., Sharma K. K., "Principles of Pharmacology", Paras medical publisher.2021
7. Marry Anne K. K., Lloyd Yee Y., Brian K. A., Robbin L.C. "Applied Therapeutics, The Clinical use of Drugs," The Lippincott Williams &Wilkins.
8. K.D.Tripathi. "Essentials of Medical Pharmacology", Jaypee Brothers Medical Publishers (P) Ltd, New Delhi.
9. Charles R.Craig & Robert ,"Modern Pharmacology with clinical Applications", Lippincott Williams and Wilkins.

24BT306P					Pharmacology of Drug Action Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To familiarize students with the principles and methodologies of experimental pharmacology using simulation software and in vitro techniques.
2. To develop hands-on skills for evaluating pharmacological effects through ethical alternatives to animal experiments.
3. To provide an understanding of the pharmacological actions of drugs on various physiological systems using computational and in vitro approaches.
4. To enhance critical thinking and problem-solving abilities through the analysis and interpretation of pharmacological data from virtual and in vitro experiments.

List of Experiments

1. Introduction to experimental pharmacology
2. To study analgesic activity by writhing test (Software Based)
3. Study of Analgesic activity with the help of "Tail Flick Apparatus" (Analgesiometer) (Software Based)
4. Study of Muscle Relaxant activity with the help of "Rota-Rod Apparatus"(Software Based)
5. Study of CNS Depressants & Stimulants Using "Actophotometer"(Software Based)
6. Study of Drugs acting on CNS (Including Anxiolytic Activity) using following modules- Elevated Plus Maze Method (Software Based)
7. The effect of insulin (hypoglycemic activity) and alloxan on blood glucose (Software Based lab)
8. To study PTZ induced convulsions in mice (Software Based)
9. Prediction of Drug toxicity using ADMET analysis (Software Based)
10. Brime Shrim Lethality Assay – *In vitro*
11. Alpha Amylase enzyme inhibition assay- *In vitro*
12. Protein albumin denaturation assay- *In vitro*

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Demonstrate proficiency in using software tools and in vitro techniques for pharmacological experiments.

CO2 : Evaluate drug activity, toxicity, and efficacy through simulations and laboratory-based assays.

CO3 : Correlate pharmacological principles with experimental outcomes obtained via ethical alternatives.

CO4 : Interpret pharmacokinetic and pharmacodynamic data using *in silico* and *in vitro* methodologies.

CO5 : Design and conduct experiments ethically, adhering to the principles of 3Rs (Replacement, Reduction, Refinement).

CO6 : Critically analyze experimental data to derive conclusions relevant to drug action and safety.

TEXT/REFERENCE BOOKS

1. Kulkarni, S. K. (2019). Handbook of Experimental Pharmacology. Vallabh Prakashan. A comprehensive resource for experimental methods in pharmacology, including simulation techniques.
2. Rang, H. P., Dale, M. M., Ritter, J. M., & Flower, R. J. (2020). Rang and Dale's Pharmacology. Elsevier. Focuses on pharmacological concepts and experimental methods.
3. OECD Guidelines for Testing Chemicals (2021). Details ethical alternatives for toxicity and pharmacological testing.
4. ADMET Predictor® User Guide. (2022). Simulations Plus. Reference guide for ADMET analysis and predictive modeling.
5. Russell, W. M. S., & Burch, R. L. (1959). The Principles of Humane Experimental Technique. Foundational text on the 3Rs for ethical experimentation.